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Started in the Window, Billed Outside It

Auditing the EV Pool-Van Charging Bay's Sunshine-Window Compliance Log Against Its Own Smart Meter

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The problem

Facilities credits a session as sunshine-window compliant the moment it starts inside the free three-hour solar window. Whether the actual charge, not just the start time, lands inside that window has gone unchecked.

Research questions

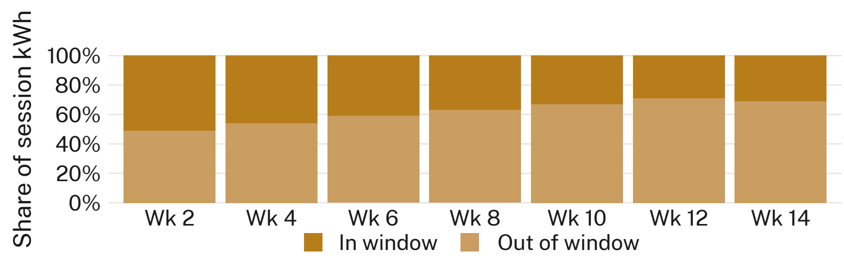
- Compliance log vs. the bay's own smart-meter readings
- Share of session kWh actually drawn inside the window
- Does start time or session length better predict that share?

Approach

- 9 pool vans · 2 bays · 5-min meter logging · 14-week term
- Session kWh reconciled against the log's compliance flag
- Charge-start held against rostered pickup, not driver discretion

Findings

The log recorded 94% of sessions as compliant. The meter attributes a median 38% of session kWh to the window, falling as the term progressed.



In-window kWh share fell from 51% in week 2 to 29% by week 12 of the 14-week deployment (9 vans, 2 bays); log-recorded compliance held above 90% throughout.

Interpretation & takeaways

A session flagged compliant at minute one can still draw most of its charge after the window shuts. The log rewards a start time; the meter prices a duration. Next: testing whether pausing the charge at window's close, not merely logging its start, moves the meter and not just the log.

Selected references

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3. Davis, B. M., & Bradley, T. H. (2012). The efficacy of electric vehicle time-of-use rates in guiding plug-in hybrid electric vehicle charging behavior. *IEEE Transactions on Smart Grid*, 3(4), 1679–1686. doi:10.1109/TSG.2012.2205951
4. Jeon, W., Cho, S., & Lee, S. (2020). Estimating the impact of electric vehicle demand response programs in a grid with varying levels of renewable energy sources: time-of-use tariff versus smart charging. *Energies*, 13(17), 4365. doi:10.3390/en13174365
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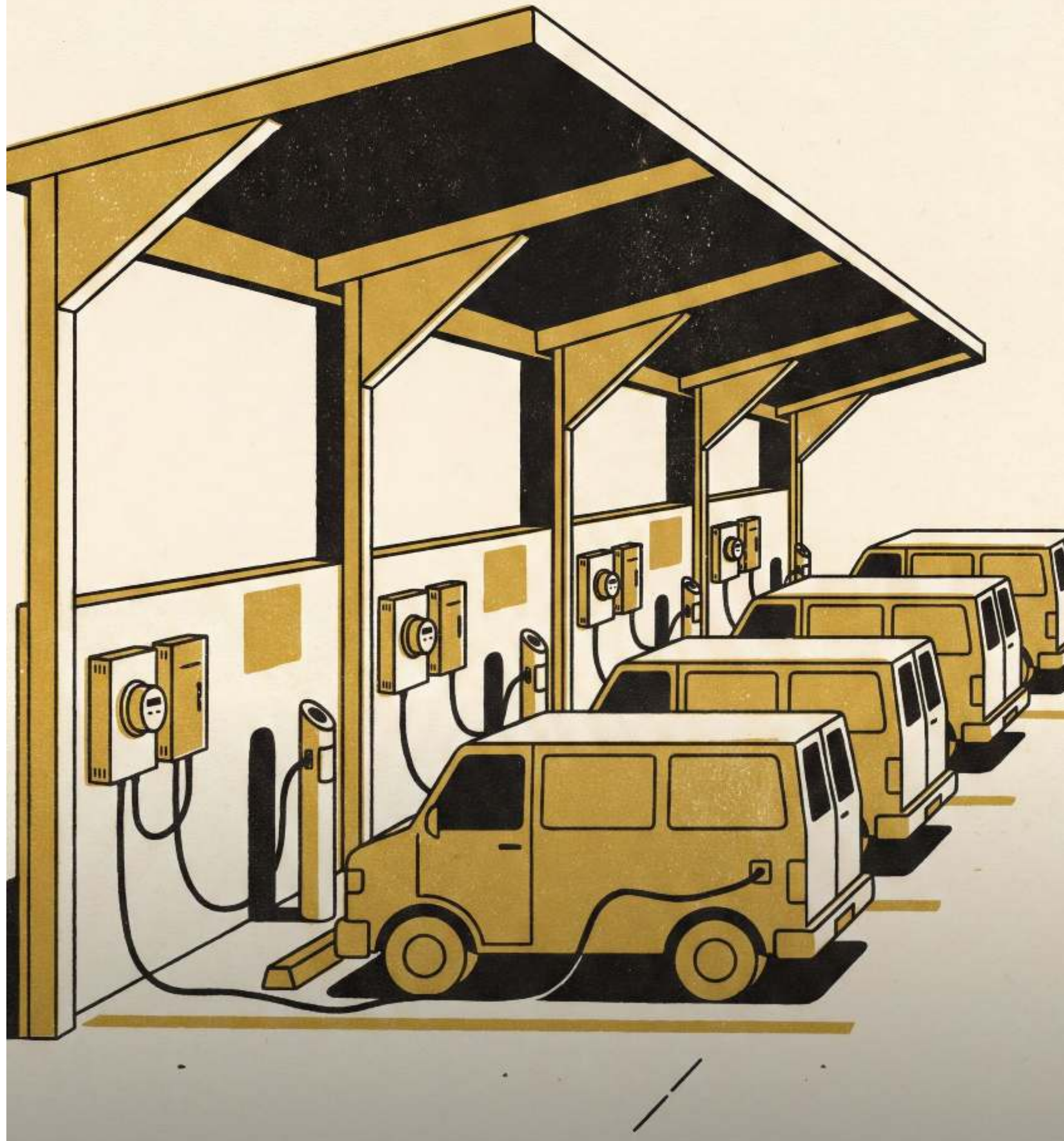
Vehicle telemetry de-identified to bay and session; no individual driver rostered against a session.



"The log says compliant. The meter says otherwise."

Median 38% of session kWh lands inside the free window across a 14-week meter audit (n = 9 vans).